Field-Level Inference with the Thomas Cluster Process

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Outline

Motivation

Field-Level NPE

Full density field → CNN → Flow

Captures all spatial information:
cluster shapes, counts, voids, higher-order correlations

Power-Spectrum MCMC

$P(k) \rightarrow$ Gaussian likelihood

Captures only two-point information:
amplitude and shape of the correlation function

Can we measure how much information the full field contains beyond $P(k)$?

Strategy: *compare two inference approaches on the same data*

Any tightening of NPE constraints over MCMC = genuine non-Gaussian information content

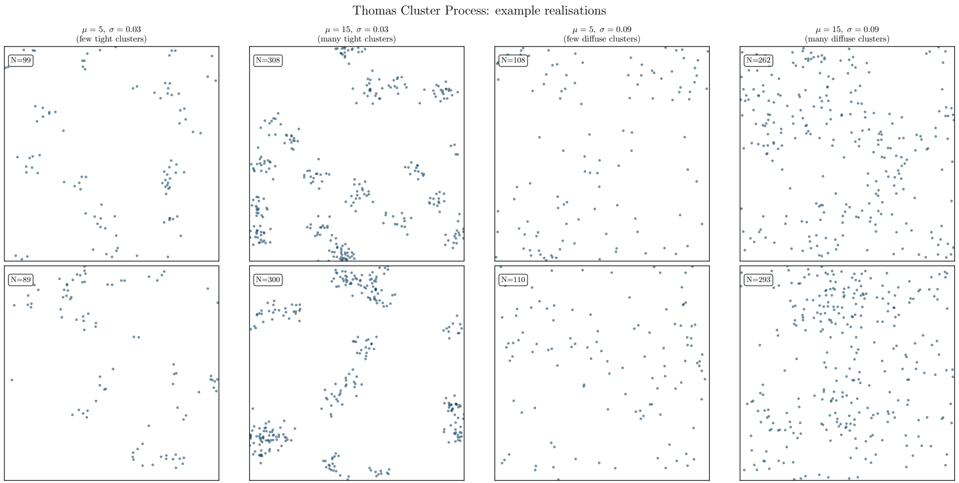
The Thomas Cluster Process

Generative model (Neyman–Scott family):

```
1. Draw N_parent ~ Poisson(λ_p = 20)
centres in [0,1]²
2. Per centre: N_j ~ Poisson(μ)
galaxies
3. Offsets: ε ~ N(0, σ²I₂), periodic
BCs
```

Free parameters: μ **(richness),** σ **(spread)**

Analogy: HOD occupation, virial radius



Priors: μ **~ U[3, 25]** σ **~ U[0.02, 0.12]**

→ ~60 to ~500 galaxies per catalog

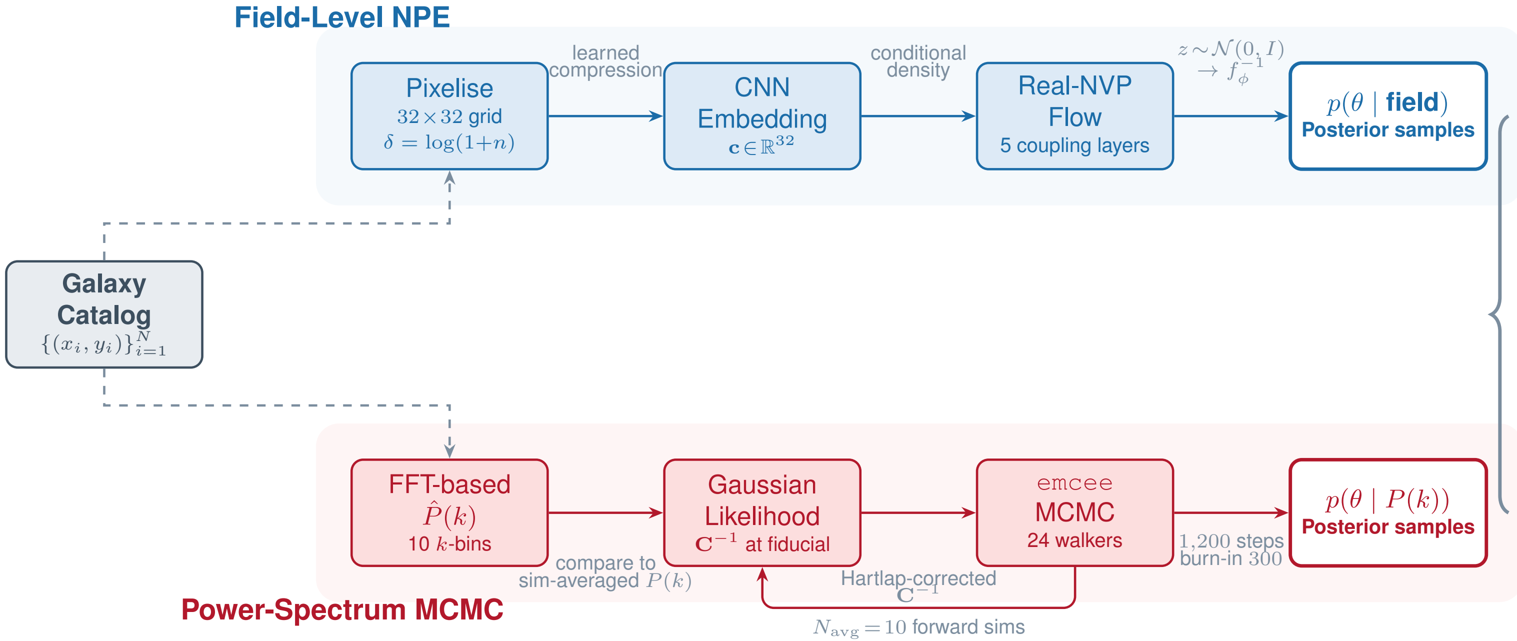
Exact closed-form: $P(k) = 1/\tilde{n} + (\mu^2/\lambda_p)exp(-k^2\sigma^2)$

where $\tilde{n} = \lambda_p\dot{\mu}$ is the mean galaxy density.

Shot noise (1/n̄)

Clustering term (amplitude × Gaussian roll-off)

Pipeline Overview: Two Inference Branches



Same observation, same simulator, same prior. Only the data representation differs.

NPE Training Loss: Negative Log-Likelihood

Forward

$$\min_{\phi} \quad \mathbb{E}_{p(y)}[\mathrm{KL}(p(\theta \mid y) \mid\mid q_{\phi}(\theta \mid y))]$$

⇓ *first term averaged over y → depends only on the true posterior, constant w.r.t.ϕ*

Expand KL

$$= \mathbb{E}_{p(\theta,y)}[\log p(\theta \mid y)] - \mathbb{E}_{p(\theta,y)}[\log q_{\phi}(\theta \mid y)]$$

Drop constant

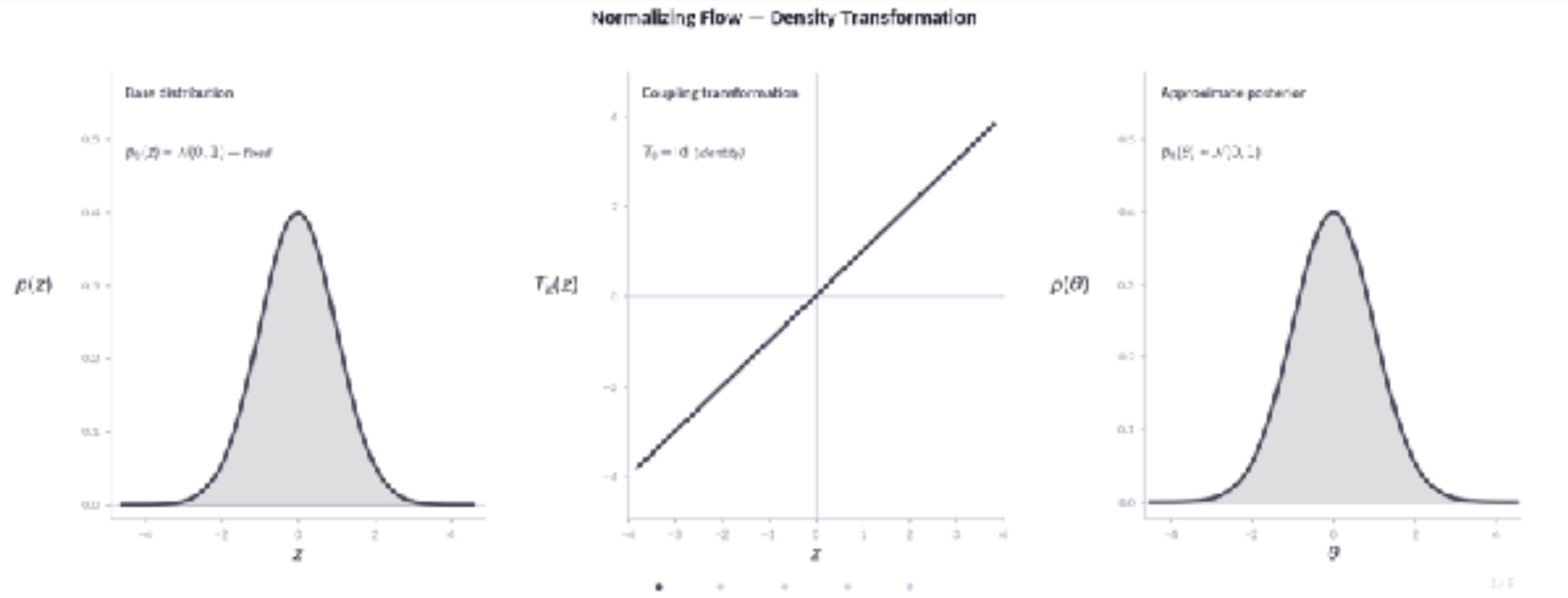
$$\min_{\phi} \mathbb{E}_{p(\theta,y)}[-\log q_{\phi}(\theta \mid y)], \quad (\theta, y) \sim p(\theta) \cdot p(y \mid \theta)$$

⇓ *Monte Carlo with N simulator draws* $\{(\theta_i, y_i)\} \sim p(\theta)p(y \mid \theta)$

Training loss

$$\mathscr{L}(\phi) = -\frac{1}{N}\sum \log q_{\phi}(\theta_i \mid y_i) \equiv \text{Negative Log-Likelihood (NLL)}$$

Normalizing Flows



Base

$$z \sim p_z(z) = \mathcal{N}(0, \mathbb{I}_d)\,.\quad z \in \mathbb{R}^d$$

Composition

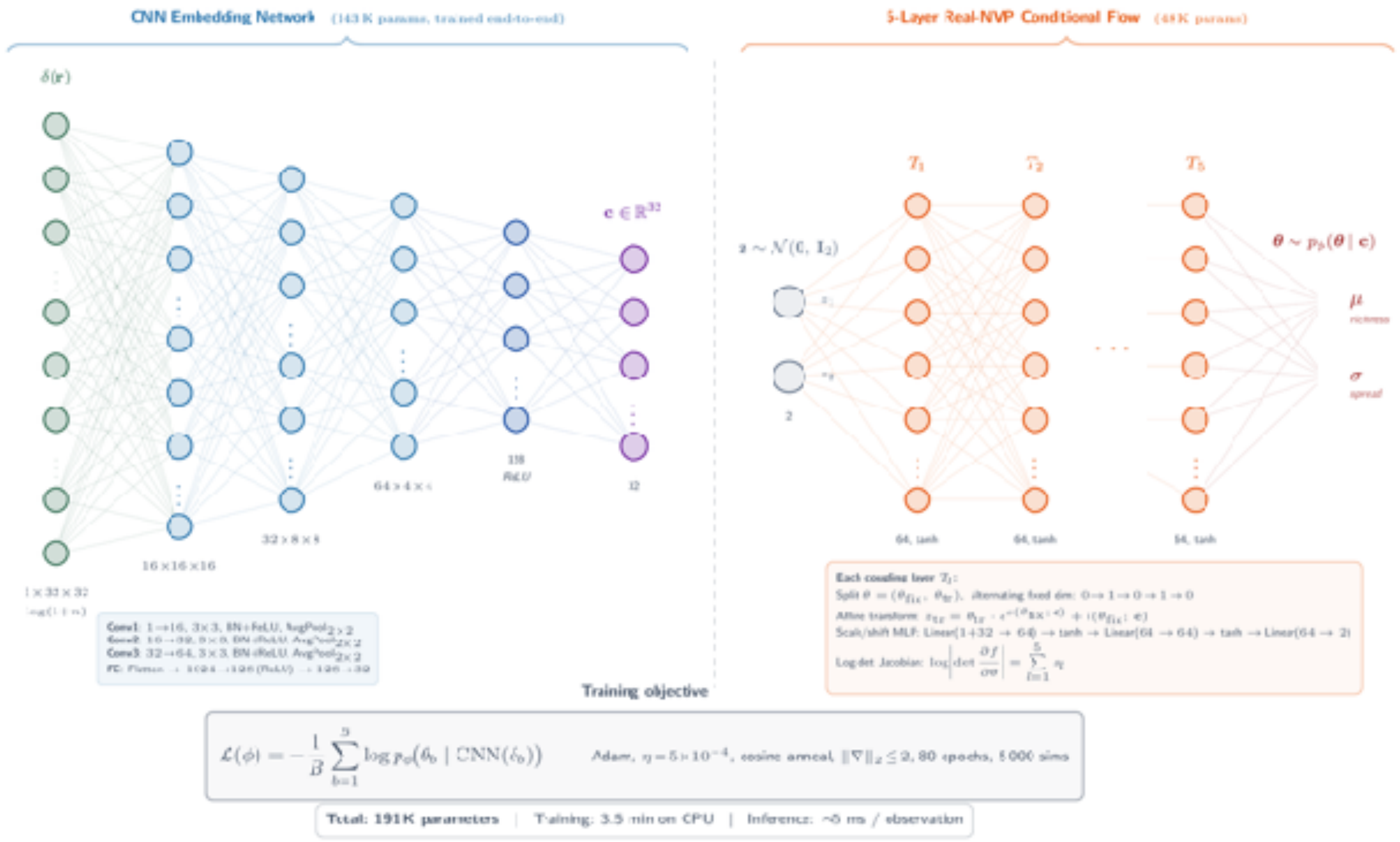
$$\theta = T(z) = (T_K \circ T_{K-1} \circ \ldots \circ T_1)(z) \quad \theta \in \mathbb{R}^d$$

Change-of-variables formula

$$\log q_\phi(\theta) = \log p_z(T^{-1}(\theta)) + \sum_{k=1}^{K} \log|detJ_{T_k^{-1}}(\theta)|$$

Requirements for T_k :

- **Invertibility:** T_k bijective, forward $T_k(z)$ and inverse
- **Tractable Jacobian:** $\log\ |\det J_{T_k}|$ computable in $O(d)$.


CNN Embedding Network (143 K params, trained end-to-end)
5-Layer Real-NVP Conditional Flow (48 K params)
$\delta(\mathbf{r})$
$1\times32\times32$
$16\times16\times16$
$32\times8\times8$
$64\times4\times4$
128
ReLU
32
$\mathbf{c}\in\mathbb{R}^{32}$
Conv1: $1\to16$, 3×3, BN+ReLU, AvgPool$_{2\times2}$
Conv3: $32\to64$, 3×3, BN+ReLU, AvgPool$_{2\times2}$
$z\sim\mathcal{N}(0,\,\mathbf{I}_2)$
2
T_1
T_2
T_5
64, tanh
64, tanh
64, tanh
$\boldsymbol{\theta}\sim p_\phi(\boldsymbol{\theta}\mid\mathbf{c})$
μ
σ
Each coupling layer T_l:
Split $\theta=(\theta_{\rm fix},\,\theta_{\rm tr})$, alternating fixed dim: $0\to1\to0\to1\to0$
Scale/shift MLP: Linear($1+32\to64$) $\to$ tanh $\to$ Linear($64\to64$) $\to$ tanh $\to$ Linear($64\to2$)
Training objective
$\mathcal{L}(\phi)=-\frac{1}{B}\sum_{b=1}^{B}\log p_\phi(\theta_b\mid\mathrm{CNN}(\delta_b))$
Total: 191K parameters | Training: 3.5 min on CPU | Inference: ~5 ms / observation

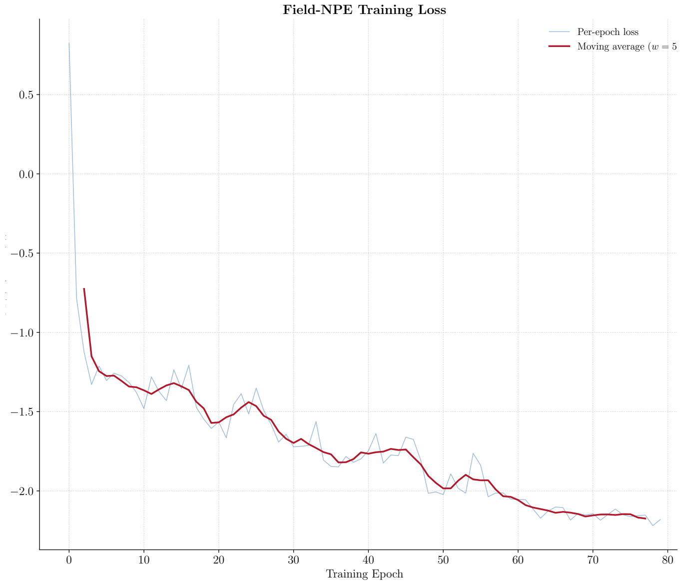
Real-NVP Flow

5 affine coupling layers, alternating fixed dimension

$$z_{tr} = \theta_{tr} \cdot \exp(s(\theta_{fix}; c)) + t(\theta_{fix}; c)$$

Scale s(·) and shift t(·) are 2-layer MLPs (64 hidden, tanh)

End-to-end training is essential: the CNN learns to extract features that minimize posterior uncertainty.



Training setup

5,000 simulations from prior

80 epochs, batch size 256

Adam, lr = 5×10^{-4}, cosine anneal

Gradient clipping $\|\nabla\|_2 \leq 2$

~3.5 min on CPU

MCMC Baseline: Power-Spectrum Likelihood

Gaussian likelihood on binned P(k):

$$\log L(\theta) = -1/2(P_{obs} - P_{model}(\theta))^{\mathrm{T}}\mathbb{C}^{-1}(P_{obs} - P_{model}(\theta))$$

- 10 $P(k)$bins via 2D FFT of the pixelized field
- 10 forward sims per likelihood evaluation (noise reduction)
- Covariance from 500 sims at fiducial + Hartlap correction
- emcee: 24 walkers × 1,200 steps (~1.5 min on CPU)

Lesson learned (the hard way):

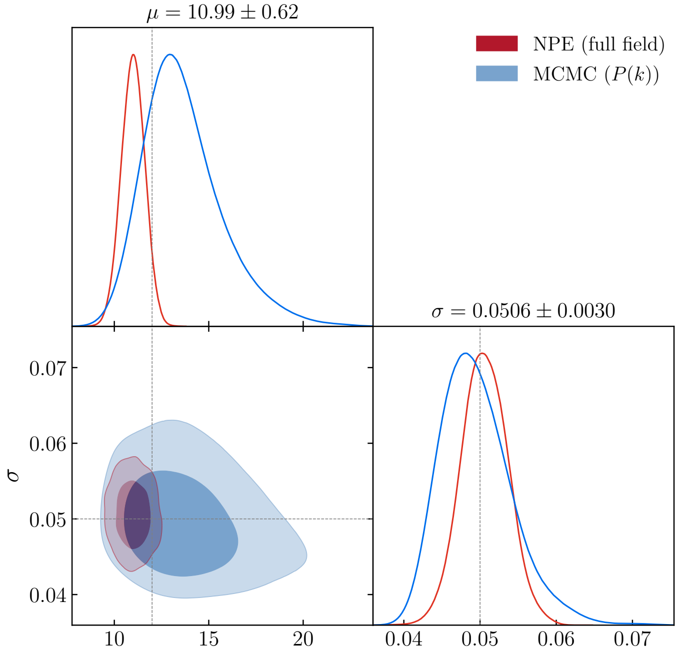
Both observed and model $P(k)$ must pass through the same FFT estimator. Comparing an analytical P(k) against an FFT estimate introduces a normalization mismatch that renders the likelihood completely uninformative — the MCMC explores the full prior.

Posterior Comparison

	True	NPE	MCMC	R
μ	12.0	**11.0 ± 0.6**	13.8 ± 2.1	**3.4×**
σ	0.050	**0.050 ± 0.003**	0.049 ± 0.004	**1.5×**

NPE contours (blue) are **visibly tighter** than **MCMC contours (red)**

R = σ_MCMC / σ_NPE = constraint-tightening ratio



Advantages

μ **: controls cluster richness, the number of galaxies in each visible cluster.**

This is a non-Gaussian, count-based feature.

$P(k)$ captures it only through the overall amplitude, the CNN can count individual clusters.

σ controls the spatial scale of clustering, the width of each cluster.

This is predominantly a Gaussian feature.

$P(k)$ captures it through the roll-off shape, so the MCMC already does well here.

For a survey with millions of observations, amortized inference is a viable path.

The CNN exploits both Gaussian and non-Gaussian features; $P(k)$ captures only the former.

Conclusions

1 **Field-level NPE outperforms P(k) MCMC**

3.4× tighter on μ, 1.5× on σ — directly quantifying non-Gaussian information content.

2 **Pixelization + CNN is simple and effective**

Permutation-invariant, computationally cheap, sufficient for spatial feature extraction.

4 **Consistent estimators matter**

Analytical vs. FFT P(k) mismatch made the MCMC likelihood uninformative.

5 **Reproducible on modest hardware**

Full pipeline in ~6 min on a single CPU core.

Future Work

F1 **Richer flow architectures**

Neural Spline Flows, MAF, etc.

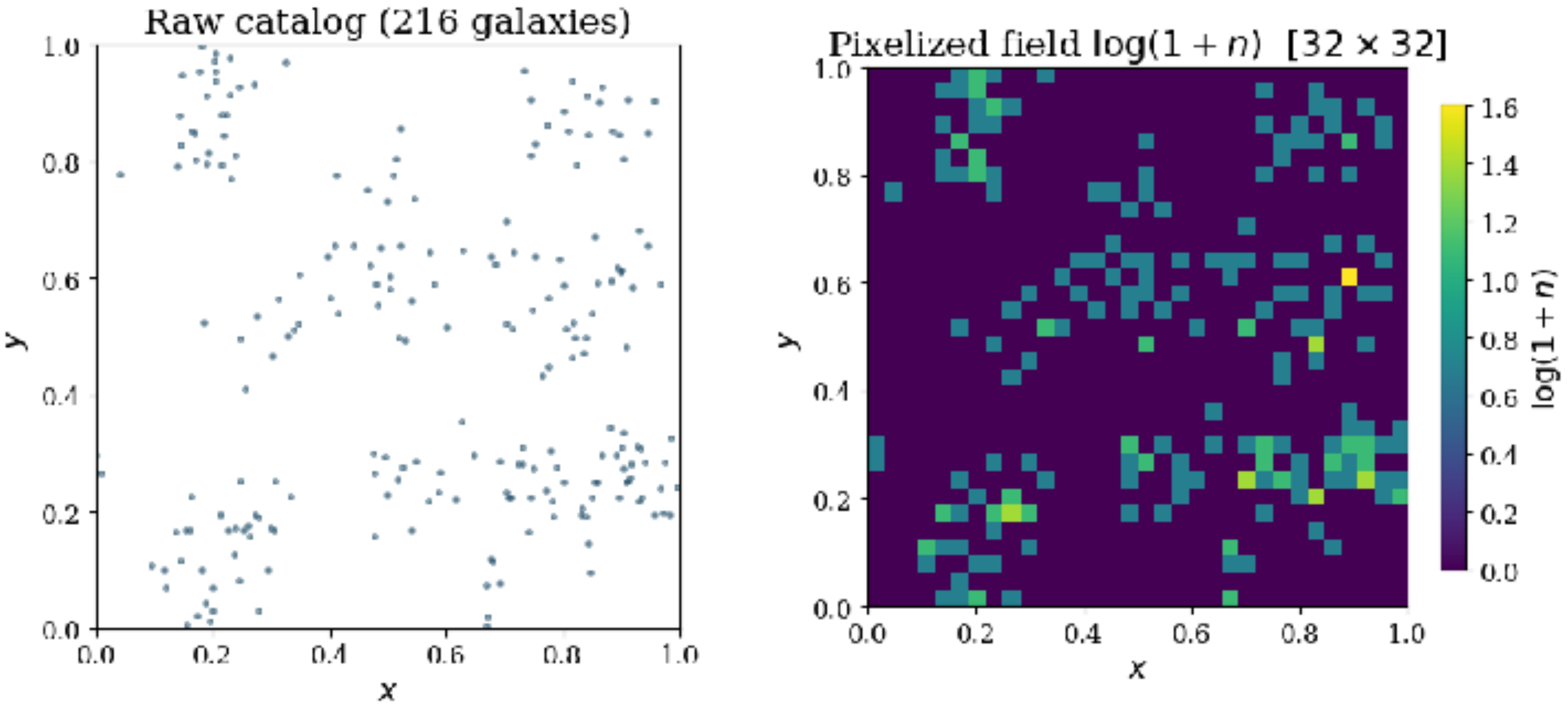
F3 **SBC validation**

Full Simulation-Based Calibration for the field-level pipeline

F4 **Three-parameter inference**

Extend to (λ_p, μ, σ) → connect to (Ω_m, σ_8) via halo model

Thank you


Raw catalog (216 galaxies)
Pixelized field log(1 + n) [32 × 32]
x
y
log(1 + n)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6